NONVANISHING LOCI ARE OPEN AND THE SECTION INVERTS THERE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1030-1043, following the statement at 1019-1028.

Editorial note. The author names the Modules engine and then gives a complete direct proof; the direct proof is the counted content.

Allowed standard prerequisites (editorial).

- Standard opens in ringed spaces. Modules, Lemma modules-lemma-s-open (cross-chapter, cited at ~1031). Named special-case engine; the direct proof is also given.

Source reference. Original reference 01CY: focused-seeds/geometry/S9-schemes-f-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a locally ringed space. Let $f \in \Gamma(X, \mathcal{O}_X)$. The set*

$$D(f) = \{x \in X \mid \text{image } f \notin \mathfrak{m}_x\}$$

is open. Moreover $f|_{D(f)}$ has an inverse.

Proof. This is a special case of Modules, Lemma 01CY, but we also give a direct proof. Suppose that $U \subset X$ and $V \subset X$ are two open subsets such that $f|_U$ has an inverse g and $f|_V$ has an inverse h. Then clearly $g|_{U\cap V} = h|_{U\cap V}$. Thus it suffices to show that f is invertible in an open neighbourhood of any $x \in D(f)$. This is clear because $f \notin \mathfrak{m}_x$ implies that $f \in \mathcal{O}_{X,x}$ has an inverse $g \in \mathcal{O}_{X,x}$ which means there is some open neighbourhood $x \in U \subset X$ so that $g \in \mathcal{O}_X(U)$ and $g \cdot f|_U = 1$. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.